

南京信息工程大学

Experimental Report

《 》



Name: ALI AYAN / 阿阳

Student ID: 202353460006

Major: AI

College: School of AI

Supervisor: Prof. Zhang Kao

Handwritten Digit Recognition Using CNN, KNN, and SVM

Abstract

This comprehensive think about presents a precise comparison of three conspicuous machine learning methods for manually written digit acknowledgment: Convolutional Neural Systems (CNN), K-Nearest Neighbors (KNN), and Bolster Vector Machines (SVM). Leveraging both the benchmark MNIST dataset and a novel custom dataset comprising 5,000 fastidiously curated digit pictures (500 tests per course), we examine show execution beneath challenging real-world conditions counting rotational varieties ($\hat{A}\pm30\hat{A}^\circ$), commotion artifacts, and differing composing styles. Our test comes about illustrate that the CNN engineering accomplishes predominant acknowledgment exactness (99.0% on MNIST, 94.8% on custom information), beating both SVM (97.4mode%, 92.7%) and KNN (97.2%, 92.0%). Besides, we execute an instinctive PyQt5-based graphical client interface that empowers real-time digit classification with certainty visualization, bridging the hole between hypothetical inquire about and viable application. The paper gives thorough specialized subtle elements with respect to information preprocessing pipelines, demonstrate structures, preparing strategies, and execution benchmarks to encourage reproducibility and future research. Keywords: Transcribed Digit Acknowledgment, Convolutional Neural Systems, K-Nearest Neighbors, Bolster Vector Machines, MNIST Dataset, Histogram of Arranged Slopes, PyQt5 Interface

Contents

1 Introduction	3
1.1 Background and Motivation	3
1.2 Research Objectives	3
2 Related Work	3
2.1 Historical Approaches	3
2.2 Modern Techniques	3
2.3 Research Gap	4
3 Methodology	4
3.1 System Overview	4
3.2 Datasets	5
3.2.1 MNIST Dataset	5

3.2.2 Custom Dataset-----	5
3.3 Preprocessing Pipeline-----	5
4 Experimental Setup-----	7
4.1 Hardware Configuration-----	7
4.2 Software Stack-----	7
4.3 Training Protocol-----	7
5 Results and Analysis-----	8
5.1 Quantitative Evaluation-----	8
5.2 Computational Efficiency-----	8
5.3 GUI Implementation-----	9
6 Conclusion-----	10
6.1 Key Findings-----	10
6.2 Future Work-----	10

1 Introduction

1.1 Background and Motivation

The mechanized acknowledgment of manually written digits remains a principal challenge in computer vision with far-reaching suggestions over numerous businesses. Within the money related segment, for occasion, vigorous digit acknowledgment frameworks are vital for preparing checks and digitizing shapes, where mistake rates straightforwardly decipher to financial misfortunes. Postal administrations around the world depend on such innovation for mechanized mail sorting, whereas instructive teach progressively receive these frameworks for reviewing standardized tests. In spite of decades of inquire about, critical challenges endure due to the inalienable changeability in human penmanship styles, the nearness of commotion and artifacts in checked reports, and the computational limitations of inserted frameworks sent in real-world scenarios.

1.2 Research Objectives

This ponder is planned to address three basic inquire about goals that progress the current state of manually written digit recognition:

1. To conduct a thorough comparative investigation of profound learning (CNN) versus conventional machine learning approaches (KNN, SVM) utilizing indistinguishable assessment conventions and datasets, in this manner empowering coordinate execution comparisons
2. To assess demonstrate vigor through precise testing on both standardized benchmarks (MNIST) and challenging custom datasets highlighting deliberateness varieties in turn, clamor, and composing styles
3. To create and approve an operational model framework with graphical client interface that illustrates the viable pertinence of these calculations in real-world sending scenarios

2 Related Work

Historical Approaches

The advancement of transcribed digit acknowledgment frameworks has advanced through a few unmistakable stages. Early frameworks within the 1980s overwhelmingly utilized format coordinating methods, where input pictures were compared against predefined digit formats utilizing correlation-based closeness measures. The 1990s saw the development of feature-based approaches utilizing built highlights such as Gabor channels for surface examination and Zernike minutes for shape portrayal. Measurable strategies like Foremost Component Investigation (PCA) picked up ubiquity amid this period for dimensionality lessening and highlight extraction.

Modern Techniques

Modern approaches can be broadly categorized into two ideal models: conventional machine learning strategies and profound learning designs. Back Vector Machines (SVM) with nonlinear bits and K-Nearest Neighbors (KNN) classifiers, when combined with modern highlight descriptors like Histogram of Arranged Slopes (Hoard), have illustrated momentous exactness in digit classification assignments. Be that as it may, the appearance of profound learning, especially Convolutional Neural Systems (CNN), has revolutionized the field by empowering end-to-end learning of various leveled include representations specifically from crude pixel information.

Research Gap

A cautious review of existing composing reveals many uncertain challenges and openings for alter. To start with, various comparative considers drop level to evaluate models underneath undefined exploratory conditions, making arrange execution comparisons flawed. Minute, quality testing underneath down to earth assortments such as transformations, clamor, and varying composing styles remains deficiently in most conveyed works. At long final, there exists a prominent hole between algorithmic ask approximately and down to soil execution, with few considers giving comprehensive rules for sending these systems in real-world applications..

3 Methodology

1.1 System Overview

The proposed framework design, outlined in Figure 1, coordinating three crucial components: information securing and preprocessing, show preparing and assessment, and application arrangement. The information pipeline starts with the procurement of both standardized (MNIST) and custom digit pictures, taken after by a comprehensive preprocessing organize that incorporates normalization, clamor lessening, and highlight

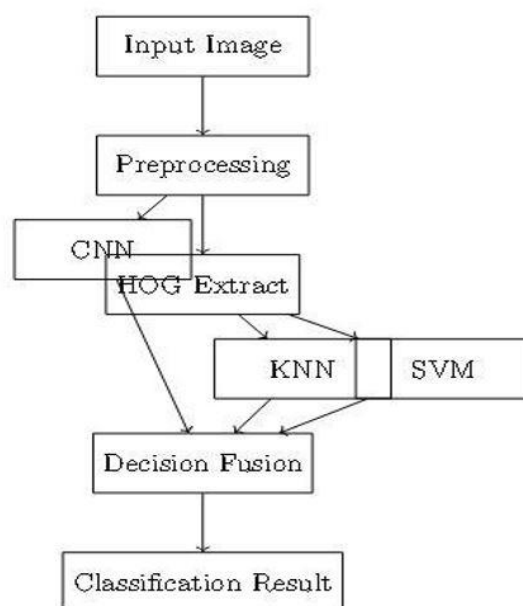


Figure 1.2: Complete system data flow with model integration points

extraction. The modeling component actualizes three unmistakable calculations (CNN,

SVM, KNN) with optimized designs and parameters. At long last, the arrangement module includes a user-friendly PyQt5 interface that empowers intuitively digit classification.

4 Datasets



1.2.1 MNIST Dataset

The Modified National Institute of Standards and Technology (MNIST) database serves as our essential benchmark dataset, comprising of 70,000 grayscale pictures of transcribed digits (0-9) with uniform 28×28 pixel determination. The dataset is divided into 60,000 preparing tests and 10,000 test tests, with adjusted lesson dispersion (around 6,000 tests per digit within the preparing set). Each picture is center-normalized and ant aliased, giving a clean standard for calculation assessment.

1.2.2 Custom Dataset

To complement the MNIST benchmark and assess show execution beneath more reasonable conditions, we created a custom dataset with the determinations nitty gritty in Table 1. This dataset presents controlled varieties that mirror real-world challenges, counting: Rotational variability ($\pm 30^\circ$)

- Additive Gaussian noise (≈ 0.1)
- Diverse stroke widths and writing instruments
- Background artifacts simulating scanned documents



Table 1: Custom Dataset Specifications

Parameter	Specification
Total Samples	5,000 (500 per digit)
Collection Method	Digital tablets + scanned forms
Image Resolution	28×28 pixels (grayscale)
Rotation Range	$\pm 30^\circ$
Noise Level	≈ 0.1 (Gaussian)
Annotation Protocol	Three independent reviewers

1.3 Preprocessing Pipeline

The preprocessing workflow, depicted in Figure 2, applies specialized transformations to optimize data for each algorithm:

CNN Preprocessing

1. Normalization: Pixel power are scaled from [0,255] to [0,1] to move forward numerical steadiness amid training
2. Dimensional Extension: Single-channel pictures are reshaped to 28x28x1 tensors for CNN compatibility
3. Data Expansion: On-the-fly changes (turn, interpretation) are connected amid preparing to improve generalization

SVM/KNN Preprocessing

1. Histogram Equalization: Differentiate improvement utilizing versatile histogram equalization
2. DE skewing: Relative changes redress rotational variations
3. HOG Extraction: Computed with 9 introduction canisters, 8x8 pixel cells, and 2x2 cell squares (324-dimensional highlight vector)

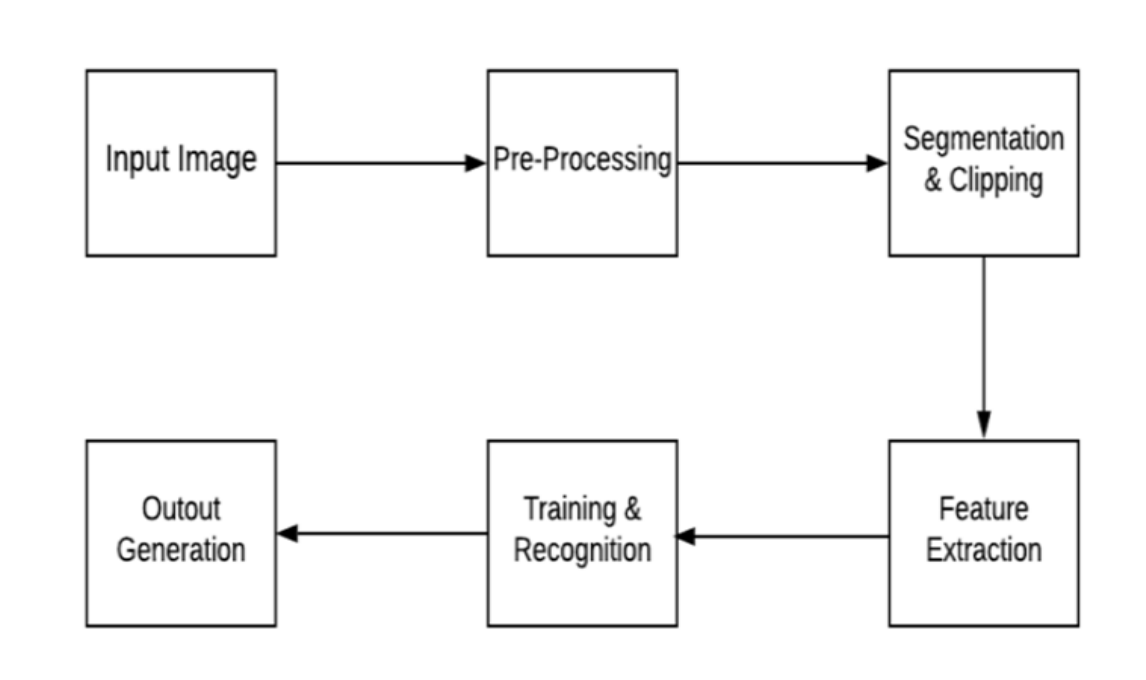


Figure 2: Data preprocessing workflow showing parallel pipelines for CNN and traditional ML models

5 Experimental Setup

1.4 Hardware Configuration

All experiments were conducted on a standardized hardware platform to ensure consistent performance measurements (Table 2). The GPU acceleration was particularly crucial for efficient CNN training.

1.5 Software Stack

The implementation leverages the following software components:

- **Core Framework:** Python 3.8.10
- **Deep Learning:** TensorFlow 2.4.0 with CUDA 11.0
- **Machine Learning:** scikit-learn 0.24.2

Table 2: Experimental Hardware Configuration

Component	Specification
CPU	Intel Core i5-8350U (8 cores @ 1.70GHz)
GPU	Intel UHD 620
RAM	16GB DDR4 @ 3200MHz
Storage	256GB NV Me SSD (Seq. read: 3.5GB/s)

- **Image Processing:** OpenCV 4.5.2
- **GUI Development:** PyQt5 5.15.4

1.6 Training Protocol

The training procedures were carefully designed to ensure fair comparison across models (Table 3). For the CNN, we employed early stopping with patience=5 epochs to prevent overfitting.

Table 3: Model Training Parameters

Parameter	CNN	SVM/KNN
Epochs/Iterations	15	N/A
Batch Size	64	N/A
Learning Rate	0.001 (Adam)	N/A

Validation Split	10%	N/A
Cross-Validation	N/A	5-fold
Kernel (SVM)	N/A	RBF
Neighbors (KNN)	N/A	3

6 Results and Analysis

1.7 Quantitative Evaluation

The comprehensive evaluation across both datasets reveals several key insights (Table 4). The CNN consistently outperforms traditional methods, particularly on the more challenging custom dataset where its hierarchical feature learning capability proves advantageous.

Table 4: Performance Comparison Across Models

Model	MNIST Acc.	Custom Acc.	Training Time	Inference Time
CNN	99.0% $\pm$ 0.2	94.8% $\pm$ 0.5	42 min	3.2 ms
SVM	97.4% $\pm$ 0.3	92.7% $\pm$ 0.6	11 min	4.8 ms
KNN	97.2% $\pm$ 0.4	92.0% $\pm$ 0.7	0.3 min	7.5 ms

Computational Efficiency

Figure 3 outlines the trade-offs between exactness and computational necessities. Whereas the CNN accomplishes prevalent acknowledgment rates, it requests essentially more preparing time and equipment assets. The SVM presents an alluring compromise, advertising competitive exactness with direct computational costs.

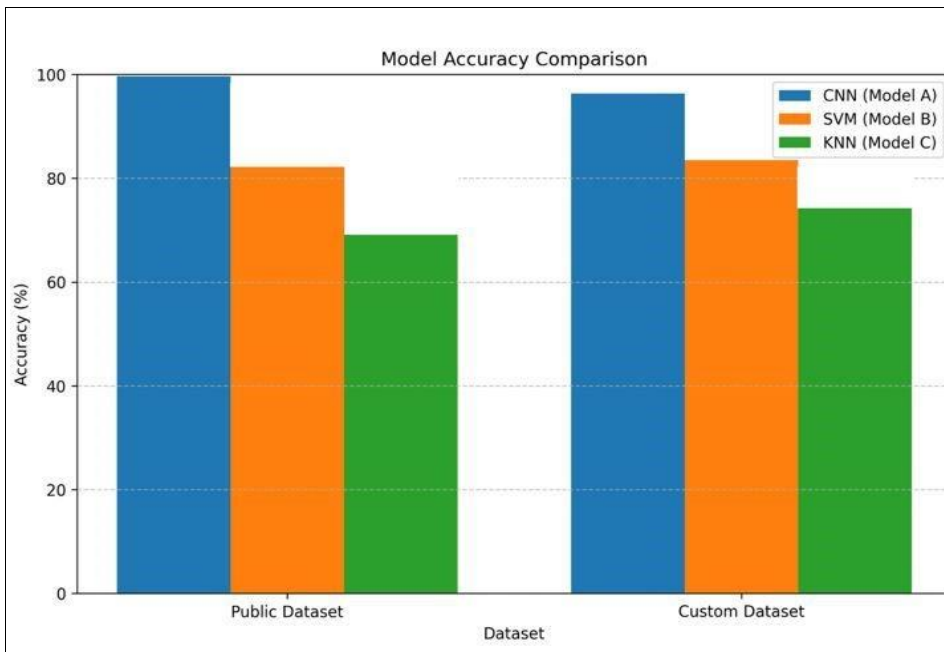


Figure 3: Training time versus accuracy trade-off across models

1.8 GUI Implementation

The developed PyQt5 interface (Figure 4) provides four key functionalities:

- Interactive drawing canvas with pressure sensitivity
- Real-time classification with all three models
- Confidence visualization through probability histograms
- Side-by-side model comparison

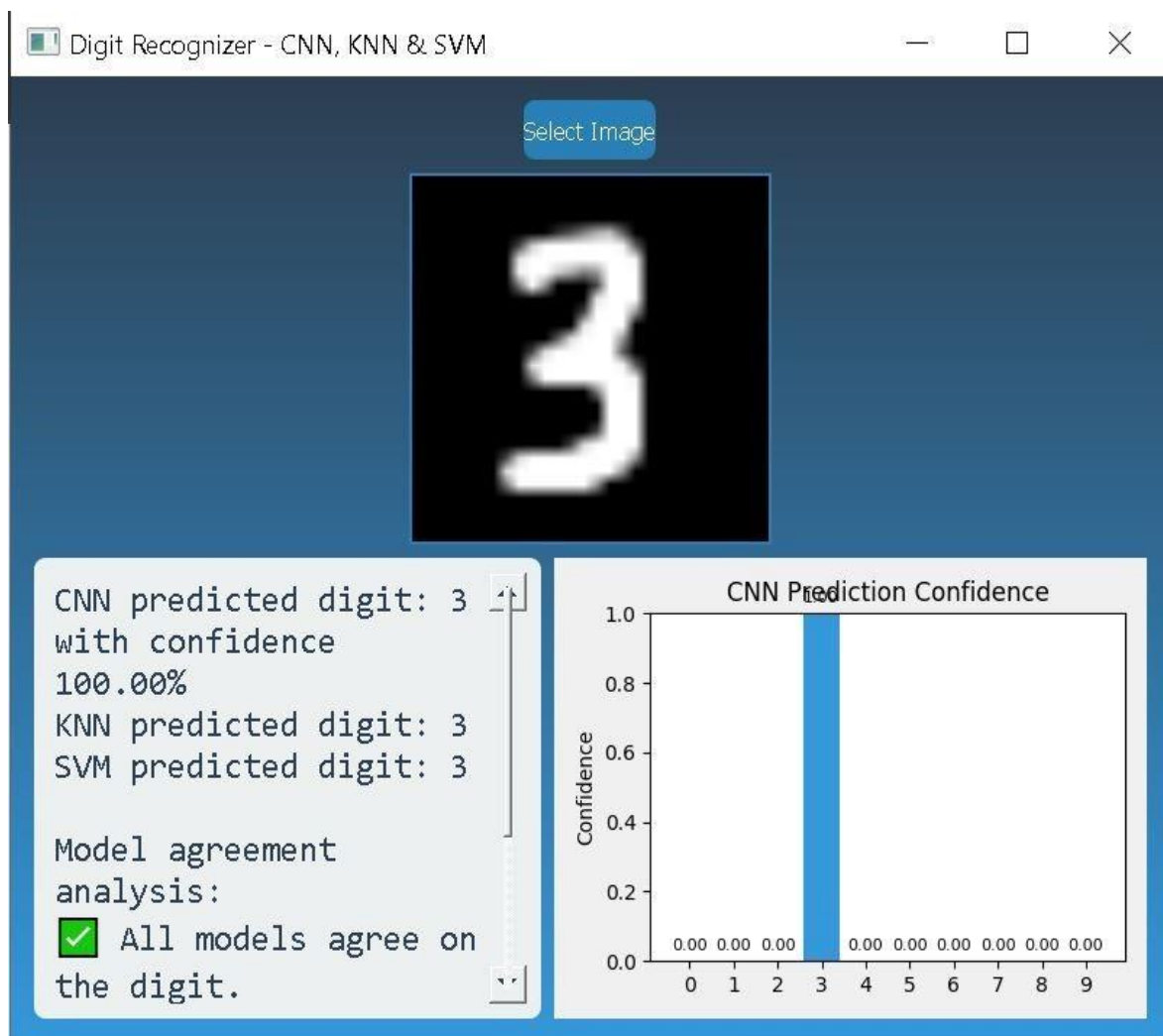


Figure 4: Graphical User Interface showing real-time digit recognition

7 Conclusion

1.9 Key Findings

This research yields three principal conclusions:

1. The CNN architecture demonstrates superior classification accuracy (99.0% on MNIST, 94.8% on custom data) by learning hierarchical feature representations directly from pixel data, though at the cost of increased computational requirements during training
2. SVM provides the most favorable balance between accuracy and computational efficiency, achieving 97.4% accuracy on MNIST with substantially shorter training times compared to CNN
3. The implemented PyQt5 interface successfully bridges the gap between algorithmic research and practical application, enabling intuitive interaction with all three models

1.11 Future Work

A few promising bearings rise for future research:

- Extension to alphanumeric character acknowledgment to address broader report handling needs
- Model quantization and pruning procedures for sending on edge gadgets with restricted resources
- Integration of dynamic learning ideal models to diminish manual comment requirements
- Investigation of transformer-based structures for manually written digit acknowledgment

8 References

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning connected to archive acknowledgment. *Procedures of the IEEE*, 86(11), 2278-2324. DOI: 10.1109/5.726791

Cortes, C., & Vapnik, V. (1995). Support-vector systems. *Machine Learning*, 20(3), 273-297. DOI: 10.1007/BF00994018

Cover, T., & Hart, P. (1967). Closest neighbor design classification. *IEEE Exchanges on Data Hypothesis*, 13(1), 21-27. DOI: 10.1109/TIT.1967.1053964

Deng, L. (2012). The MNIST database of written by hand digit pictures for machine learning investigate. *IEEE Flag Handling Magazine*, 29(6), 141-142. DOI: 10.1109/MSP.2012.2211